

Math Made Clear

Optional Subtitle

Author: Jueon Kim, Yeoeun Kim



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About the Author

Short author bio here. (e.g., background and inspiration; include website/contact if desired.) Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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Preface

Mathematics is a universal language, but learning it in a second language can be both a challenge and an opportunity. This book was created especially for Korean students who are studying math in English. Our purpose is to bridge the gap between two worlds—helping you master essential math concepts while also building your confidence in English.

The motivation for this book comes from the experiences of many students who have struggled to understand math vocabulary, instructions, and problem-solving strategies in English. We know that with the right support, every student can succeed. By providing clear explanations, bilingual vocabulary, and step-by-step examples, we hope to make math accessible and enjoyable for you.

Whether you are preparing for international exams, planning to study abroad, or simply want to strengthen your math and language skills, this book is here to support your journey. Remember, every great mathematician started with curiosity and perseverance. We hope this book inspires you to explore, ask questions, and discover the beauty of mathematics in both Korean and English.

Introduction

Welcome to this math book designed especially for Korean students learning English! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Geometry, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with yourself and celebrate your progress!

We hope this book becomes a valuable companion on your journey to mastering math and English together.

Chapter 1

Numbers & Operations

Welcome! In this book, you'll explore the world of numbers and how we use them every day. You'll learn about place value, decimals, fractions, and more. Each Section includes explanations, examples, and practice problems to help you master these important math skills.

1.1 Place Value

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together.

We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

“In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number.”

1.1.1 What is Place Value?

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

To truly grasp the concept of place value, for example, let’s take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit’s value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let’s see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In

another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

“Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.”

Short Question (English)

What is the value of the digit 8 in the number 9,836?

Short Question (Korean)

What is the value of the digit 8 in the number 9,836?

Short Question (Answer)

What is the value of the digit 8 in the number 9,836?

Answer: 800 — it's in the **hundreds place**.

Long Question (English)

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

Long Question (Korean)

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

Long Question (Answer)

Write the number 5,041 in expanded form. Then explain what each digit means based on its place.

Answer:

Expanded form: $5000 + 0 + 40 + 1$

Explanation:

- $5 = 5000$ (thousands place)
- $0 = 0$ (hundreds place)
- $4 = 40$ (tens place)
- $1 = 1$ (ones place)

Challenge Question (English)

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

Challenge Question (Korean)

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

Challenge Question (Answer)

Two numbers have the same digits but in a different order: 5,104 and 4,105. Which number is greater, and how do you know? Explain using place value.

Answer:

5,104 is greater because the **5** is in the **thousands place**, meaning 5000.

In 4,105, the **4** is in the thousands place, meaning 4000.

So: **$5,104 > 4,105$**

1.2 Decimals and Fractions

Sometimes, a whole number is too much. We want to describe something in parts: half a sandwich, a third of a mile, seventy-five hundredths of a dollar. That's when we use fractions and decimals. There are two ways of saying the same thing: less than one. Fractions use numerators and denominators. Decimals use place value. But both express parts of a whole.

Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$. The fraction $\frac{3}{10}$ becomes 0.3. The decimal 0.6 becomes $\frac{6}{10}$, which is $\frac{3}{5}$. Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.

“Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$.”

1.2.1 What are Decimals and Fractions?

As we move beyond whole numbers, we encounter quantities that are less than one. These are often represented as fractions or decimals. A **fraction** shows a part of a whole, such as $\frac{1}{2}$ or $\frac{3}{4}$. On the other hand, a **decimal** expresses the same idea using a decimal point, like 0.5 or 0.75. Both forms are just different ways to describe the same concept.

For example, consider the decimal 0.75. This can be written as the fraction $\frac{75}{100}$, which simplifies to $\frac{3}{4}$.

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Let's try converting between these forms. If you have the fraction $\frac{3}{10}$, you can write it as the decimal 0.3. Conversely, if you start with the decimal 0.6, you can express it as a fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$ in simplest form. Being able to move between decimals and fractions is a key skill that will help you in many areas of math.

“Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.”

Short Question (English)

Write the fraction $\frac{3}{10}$ as a decimal.

Short Question (Korean)

Write the fraction $\frac{3}{10}$ as a decimal.

Short Question (Answer)

Write the fraction $\frac{3}{10}$ as a decimal.

Answer: 0.3 — Divide 3 by 10 to get the decimal.

Long Question (English)

Convert the decimal 0.6 into a simplified fraction. Then explain each step.

Long Question (Korean)

Convert the decimal 0.6 into a simplified fraction. Then explain each step.

Long Question (Answer)

Convert the decimal 0.6 into a simplified fraction. Then explain each step.

Answer:

Step 1: Write 0.6 as $\frac{6}{10}$

Step 2: Simplify by dividing top and bottom by 2: $\frac{6}{10} = \frac{3}{5}$

Answer: $\frac{3}{5}$

Challenge Question (English)

Emma says that the fraction $\frac{4}{5}$ is equal to the decimal 0.45. Is she correct? Explain your reasoning.

Challenge Question (Korean)

Emma says that the fraction $\frac{4}{5}$ is equal to the decimal 0.45. Is she correct? Explain your reasoning.

Challenge Question (Answer)

Emma says that the fraction $\frac{4}{5}$ is equal to the decimal 0.45. Is she correct? Explain your reasoning.

Answer:

No, because $\frac{4}{5} = 0.8$, not 0.45.

$0.45 = \frac{9}{20}$, not the same as $\frac{4}{5}$.

1.3 Comparing and Ordering Numbers

Math is not just about knowing what numbers are. It is about knowing how they relate. We compare. We rank. We ask, which is bigger? Which is smaller? Which is equal? Symbols like $>$, $<$, and $=$ help us answer. But first, we have to understand the values.

Take 5.03 and 5.3. They may look close. But 5.3 is larger. Now try comparing fractions: $\frac{2}{5}$ and $\frac{3}{7}$. Convert them to decimals. $\frac{2}{5}$ is 0.4. $\frac{3}{7}$ is about 0.428. So $\frac{3}{7}$ is greater. This kind of comparison matters in real life—from prices to measurements to data.

Ordering numbers builds the same skill. Imagine you are given 0.2, $\frac{1}{3}$, 0.25, and 0.199. Which is the greatest? Which is least? Convert, compare, and you find the order. This process teaches more than math. It teaches precision, logic, and clarity.

“Math is not just about knowing what numbers are. It is about knowing how they relate.”

1.3.1 How do we compare and order numbers?

In mathematics, it's important not only to understand numbers, but also to compare them and put them in order. We use symbols like $>$ (greater than), $<$ (less than), and $=$ (equal to) to show how numbers relate to each other. For example, if you are given two numbers, you can decide which is larger or smaller, or if they are the same.

Let's look at a few examples. Suppose you want to compare 5.03 and 5.3. Since $5.03 < 5.3$, we use the less than symbol. Now, what if you have two fractions, $\frac{2}{5}$ and $\frac{3}{7}$? By converting them to decimals, you find that $\frac{2}{5} = 0.4$ and $\frac{3}{7} \approx 0.428$, so $0.4 < 0.428$, which means $\frac{2}{5} < \frac{3}{7}$.

Ordering numbers is also a useful skill. Imagine you are asked to arrange the numbers 0.2, $\frac{1}{3}$, 0.25, 0.199 from greatest to least. By converting them all to decimals, you see that $\frac{1}{3}$ (approximately 0.333) is the largest, followed by 0.25, then 0.2, and finally 0.199. Sometimes, you may need to compare a fraction and a decimal, such as $\frac{4}{9}$ and 0.44. Since $\frac{4}{9} \approx 0.444$, it is slightly greater than

0.44. These skills are essential for working with numbers in real-life situations.

“Take 5.03 and 5.3. They may look close. But 5.3 is larger.”

Short Question (English)

Which is greater: 5.03 or 5.3?

Short Question (Korean)

Which is greater: 5.03 or 5.3?

Short Question (Answer)

Which is greater: 5.03 or 5.3?

Answer: 5.3 is greater — it has more value in the tenths place.

Long Question (English)

Put the following numbers in order from least to greatest: 0.199 , $\frac{1}{3}$, 0.25 , 0.2

Long Question (Korean)

Put the following numbers in order from least to greatest: 0.199 , $\frac{1}{3}$, 0.25 , 0.2

Long Question (Answer)

Put the following numbers in order from least to greatest: 0.199 , $\frac{1}{3}$, 0.25 , 0.2

Answer:

Convert to decimals:

- 0.199

- $\frac{1}{3} \approx 0.333$

- 0.25

- 0.2

Ordered: 0.199 ; 0.2 ; 0.25 ; $\frac{1}{3}$

Challenge Question (English)

Which is greater: $\frac{4}{9}$ or 0.44? Show your reasoning.

Challenge Question (Korean)

Which is greater: $\frac{4}{9}$ or 0.44? Show your reasoning.

Challenge Question (Answer)

Which is greater: $\frac{4}{9}$ or 0.44? Show your reasoning.

Answer:

$\frac{4}{9} \approx 0.444...$ which is slightly greater than 0.44.

Answer: $\frac{4}{9}$ is greater.

1.4 Factors and Multiples

Some numbers fit together cleanly. Others do not. Factors and multiples explain why. A factor is a number that divides evenly. A multiple is what you get when you multiply. These two ideas help us break numbers apart and build them back up.

If you list the factors of 12, you get 1, 2, 3, 4, 6, and 12. Each divides 12 without a remainder. The multiples of 5 are 5, 10, 15, 20, and so on. These ideas may seem mechanical, but they are powerful. They lay the groundwork for division, fractions, and algebra.

“A factor is a number that divides evenly. A multiple is what you get when you multiply.”

1.4.1 What are Factors and Multiples?

As you continue your journey in mathematics, you'll often need to break numbers down or build them up. This is where factors and multiples come in. A **factor** is a number that divides another number exactly, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12, because each of these divides 12 evenly. On the other hand, a **multiple** is what you get when you multiply a number by an integer. For instance, the multiples of 5 are 5, 10, 15, 20, and so on.

Let's see how this works with a few examples. If you want to find all the factors of 24, you list out 1, 2, 3, 4, 6, 8, 12, and 24. If you're asked for the first 6 multiples of 8, you simply multiply: 8, 16, 24, 32, 40, and 48. Sometimes, you might wonder if one number is a factor of another. For example, is 9 a factor of 81? Since $81 \div 9 = 9$ with no remainder, the answer is yes. Understanding factors and multiples is crucial for topics like division, fractions, and finding patterns in numbers.

Short Question (English)

List all the factors of 18.

Short Question (Korean)

List all the factors of 18.

Short Question (Answer)

List all the factors of 18.

Answer: 1, 2, 3, 6, 9, 18

Long Question (English)

What are the first six multiples of 7? Explain how you found them.

Long Question (Korean)

What are the first six multiples of 7? Explain how you found them.

Long Question (Answer)

What are the first six multiples of 7? Explain how you found them.

Answer:

7, 14, 21, 28, 35, 42 — Multiply 7 by 1 through 6.

Challenge Question (English)

Is 9 a factor of 81? Is 81 a multiple of 9? Explain.

Challenge Question (Korean)

Is 9 a factor of 81? Is 81 a multiple of 9? Explain.

Challenge Question (Answer)

Is 9 a factor of 81? Is 81 a multiple of 9? Explain.

Answer:

Yes. $81 \div 9 = 9$ — so 9 is a factor, and 81 is a multiple.

1.5 Prime and Composite Numbers

Some numbers stand alone. Others are made of parts. Prime numbers have only two factors: 1 and themselves. Composite numbers have more. This is not just a label—it is a clue to how numbers work.

2, 3, 5, 7, 11, 13, and 17 are prime. They cannot be broken down. But 15 is composite: it splits into 3 and 5. Learning to recognize prime and composite numbers helps us factor numbers, simplify fractions, and understand the building blocks of math.

“Prime numbers have only two factors: 1 and themselves.
Composite numbers have more.”

1.5.1 What are Prime and Composite Numbers?

As you explore numbers further, you’ll discover that some numbers have only two factors, while others have many. A **prime number** is a number greater than 1 that has exactly two distinct factors: 1 and itself. For example, 2, 3, 5, 7, 11, 13, 17, and 19 are all prime numbers under 20. In contrast, a **composite number** has more than two factors. For instance, 15 is composite because it can be divided by 1, 3, 5, and 15.

Let’s practice identifying these. Consider the numbers 14, 17, 21, 23, and 25. Among these, 17 and 23 are prime because they have only two factors. The others—14, 21, and 25—are composite. If you look at all the composite numbers between 10 and 20, you’ll find 10, 12, 14, 15, 16, 18, and 20. It’s also important to note that the number 1 is neither prime nor composite, since it has only one factor: itself. Recognizing prime and composite numbers is a foundational skill for understanding more advanced math concepts like prime factorization and divisibility.

Short Question (English)

Is 11 a prime or composite number?

Short Question (Korean)

Is 11 a prime or composite number?

Short Question (Answer)

Is 11 a prime or composite number?

Answer: Prime — only two factors: 1 and 11.

Long Question (English)

Identify all the prime numbers between 10 and 20. Explain how you know.

Long Question (Korean)

Identify all the prime numbers between 10 and 20. Explain how you know.

Long Question (Answer)

Identify all the prime numbers between 10 and 20. Explain how you know.

Answer:

11, 13, 17, 19 — they each have exactly two factors: 1 and themselves.

Challenge Question (English)

Why is the number 1 neither prime nor composite?

Challenge Question (Korean)

Why is the number 1 neither prime nor composite?

Challenge Question (Answer)

Why is the number 1 neither prime nor composite?

Answer:

It has only **one factor** (itself), not two or more. So it's **neither**.

1.6 GCM & LCM

Sometimes two numbers meet in the middle. The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple. These ideas sound opposite, but they both help us solve problems with shared values.

Take 12 and 18. Their GCF is 6 since it is the biggest number that divides them both. Their LCM is 36 which is the smallest number they both fit into. We find these using prime factorization. Once you learn how, you see how numbers connect. And once you see the connection, math becomes clearer.

“The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple.”

1.6.1 What are GCF and LCM?

When working with two or more numbers, it's often useful to find numbers they have in common. The **Greatest Common Factor (GCF)** is the largest number that divides two numbers exactly, while the **Least Common Multiple (LCM)** is the smallest number that both numbers divide into evenly.

To find the GCF and LCM, you can use prime factorization. For example, let's find the GCF and LCM of 12 and 18. The prime factorization of 12 is $2^2 \times 3$, and for 18 it's 2×3^2 . The GCF is the product of the lowest powers of common primes: $2^1 \times 3^1 = 6$. The LCM is the product of the highest powers: $2^2 \times 3^2 = 36$.

Let's apply this to a few more examples. The GCF of 24 and 36 is 12, since 12 is the largest number that divides both. The LCM of 8 and 12 is 24, which is the smallest number both can divide into. For 14 and 28, we see that $14 = 2 \times 7$ and $28 = 2^2 \times 7$, so the GCF is 14 and the LCM is 28. Mastering GCF and LCM will help you with fractions, ratios, and solving many real-world problems.

Short Question (English)

What is the GCF of 8 and 12?

Short Question (Korean)

What is the GCF of 8 and 12?

Short Question (Answer)

What is the GCF of 8 and 12?

Answer: 4 — the largest number that divides both.

Long Question (English)

Find the LCM of 6 and 10 using prime factorization. Show your steps.

Long Question (Korean)

Find the LCM of 6 and 10 using prime factorization. Show your steps.

Long Question (Answer)

Find the LCM of 6 and 10 using prime factorization. Show your steps.

Answer:

- $6 = 2 \times 3$

- $10 = 2 \times 5$

Take the highest powers: $2 \times 3 \times 5 = \mathbf{30}$

Challenge Question (English)

The GCF of two numbers is 7. One number is 35. What are possible values of the other number?

Challenge Question (Korean)

The GCF of two numbers is 7. One number is 35. What are possible values of the other number?

Challenge Question (Answer)

The GCF of two numbers is 7. One number is 35. What are possible values of the other number?

Answer:

The other number must be divisible by 7 but share no higher factor with 35.

Examples: 7, 14, 21, 28, 49, etc.

1.7 Tip for Students

Math is not just about solving problems. It is about seeing how ideas fit together. Learn the basics—place value, fractions, comparisons, factors, primes, and multiples. They are not just chapters. They are tools. And the better you understand them, the more confidently you can use them anywhere.

Interesting Facts

- The word “digit” comes from the Latin word for finger, because people used their fingers to count.
- The decimal system is called “base-10” because each place is ten times the value of the place to its right.
- The largest single-digit number in the decimal system is 9.
- The introduction of zero by Indian mathematicians made calculations much clearer and easier.

Chapter 2

Expressions & Equations

This is an example sentence with an index entry. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum.

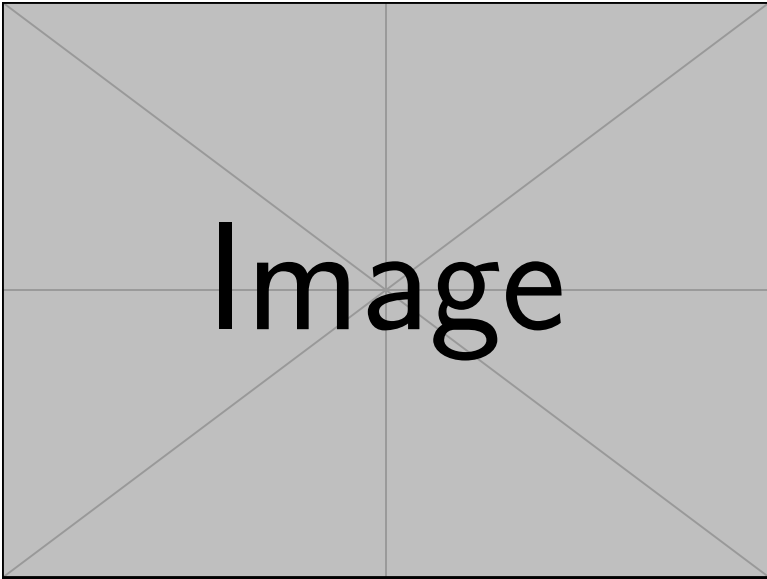


Figure 2.1: A scenic travel destination (example image).

Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

2.1 Variables and Algebraic Expressions

Welcome to the world of algebra! In this section, we'll learn how to use letters like x and y to represent numbers. These letters are called **variables**, and they help us write expressions and solve equations. Algebra might seem like a new language, but once you understand the rules, it can actually make math easier.

A **variable** is a symbol, like x or y , that stands for a number. We don't always know what the number is right away, but we can still work with it. An **algebraic expression** is a mix of numbers, variables, and math operations such as $+$, $-$, $\times$, or $\div$.

Example: Understanding Variables and Expressions

Consider the expression $3x + 5$. This means "3 times some number, then add 5." If $x = 2$, then $3x + 5 = 3(2) + 5 = 6 + 5 = 11$. This is how algebra works: we write ideas using letters and numbers, then solve them by thinking logically.

Imagine you earn \$10 per hour. If h is the number of hours you work, then your earnings are written as $10h$. If you work 5 hours, that's $10(5) = 50$. Variables help us model real-life situations.

Think of a variable as a mystery box. What number could fit inside? Try different values to see how expressions change. When writing expressions, use colors, arrows, or circles to match terms and operations.

The word "**algebra**" comes from the Arabic term "**al-jabr**," which means "reunion of broken parts." It appeared in a famous math book written over 1,200 years ago by a scholar named Al-Khwarizmi. Many English math words like "algorithm" also come from his name!

Short Question (English)

If $x = 3$, find the value of $2x + 7$.

Short Question (Korean)

If $x = 3$, find the value of $2x + 7$.

Short Question (Answer)

If $x = 3$, find the value of $2x + 7$.

Answer: $2(3) + 7 = 6 + 7 = 13$

Long Question (English)

Write an expression for "five less than triple a number y " and evaluate it when $y = 4$.

Long Question (Korean)

Write an expression for "five less than triple a number y " and evaluate it when $y = 4$.

Long Question (Answer)

Write an expression for "five less than triple a number y " and evaluate it when $y = 4$.

Answer:

Expression: $3y - 5$

When $y = 4$: $3(4) - 5 = 12 - 5 = \mathbf{7}$

Challenge Question (English)

The cost of renting a bicycle is \$8 plus \$3 per hour. Write an expression for the total cost if you rent it for h hours. Find the cost if you rent it for 5 hours.

Challenge Question (Korean)

The cost of renting a bicycle is \$8 plus \$3 per hour. Write an expression for the total cost if you rent it for h hours. Find the cost if you rent it for 5 hours.

Challenge Question (Answer)

The cost of renting a bicycle is \$8 plus \$3 per hour. Write an expression for the total cost if you rent it for h hours. Find the cost if you rent it for 5 hours.

Answer:

Expression: $8 + 3h$

If $h = 5$: $8 + 3(5) = 8 + 15 = \mathbf{23}$ dollars

2.2 Simplifying Expressions

In this section, we'll learn how to combine like terms and use the distributive property to make expressions cleaner and easier to work with. Simplifying expressions helps make your work cleaner and your thinking clearer. It's like cleaning up a messy desk so you can find what you need.

Like terms have the same variable and exponent. You can add or subtract them just like regular numbers. The **distributive property** lets you spread a number across parentheses:

$$a(b + c) = ab + ac$$

Example: Combining Like Terms and Distribution

Let's look at some examples of simplifying expressions:

Combining Like Terms:

$$2x + 3x = 5x$$

Using the Distributive Property:

$$3(x + 4) = 3x + 12$$

Whether you're solving equations or designing a budget, simplification makes everything easier. Even NASA engineers and video game developers simplify expressions when building models and writing code. It's a skill that goes far beyond school!

When simplifying, circle or highlight like terms before you combine them. Be careful with minus signs—they belong to the number or variable that follows! Go one step at a time to avoid careless mistakes.

Short Question (English)

Simplify: $7x + 5x$

Short Question (Korean)

Simplify: $7x + 5x$

Short Question (Answer)

Simplify: $7x + 5x$

Answer: $12x$

Long Question (English)

Use the distributive property to simplify $4(a + 6)$.

Long Question (Korean)

Use the distributive property to simplify $4(a + 6)$.

Long Question (Answer)

Use the distributive property to simplify $4(a + 6)$.

Answer: $4a + 24$

Challenge Question (English)

Simplify: $3x + 7 - 2x + 4$

Challenge Question (Korean)

Simplify: $3x + 7 - 2x + 4$

Challenge Question (Answer)

Simplify: $3x + 7 - 2x + 4$

Answer: Combine like terms: $(3x - 2x) + (7 + 4) = x + 11$

2.3 Solving One-Step Equations

In this section, we'll learn how to solve equations that can be finished in just one step. We'll use **inverse operations** to isolate the variable. That means undoing what's being done to the variable. Always **do the same thing to both sides** of the equation to keep it balanced.

Solving equations is a lot like solving a riddle or puzzle. You're looking for the one number that makes the equation true. Use inverse operations to isolate the variable. That means undoing what's being done to the variable.

Example: Using Inverse Operations

Let's look at some examples of solving one-step equations:

Addition Example:

$$x + 4 = 9 \quad \text{Subtract 4 from both sides} \quad x = 5$$

Multiplication Example:

$$5x = 20 \quad \text{Divide both sides by 5} \quad x = 4$$

Let's say you have \$20, and you know that it's 5 times the cost of one notebook. You can write $5x = 20$ and solve for x to find out the price of one notebook. Algebra helps you work backward to find unknowns.

When solving equations, ask: "What's happening to the variable?" Then do the opposite. Keep both sides of the equation equal—whatever you do to one side, do to the other. Always plug your answer back in to make sure it works!

Short Question (English)

Solve: $x - 9 = 15$

Short Question (Korean)

Solve: $x - 9 = 15$

Short Question (Answer)

Solve: $x - 9 = 15$

Answer: $x = 24$

Long Question (English)

Solve: $6x = 54$

Long Question (Korean)

Solve: $6x = 54$

Long Question (Answer)

Solve: $6x = 54$

Answer: $x = 9$

Challenge Question (English)

Solve: $\frac{x}{8} = 6$

Challenge Question (Korean)

Solve: $\frac{x}{8} = 6$

Challenge Question (Answer)

Solve: $\frac{x}{8} = 6$

Answer: Multiply both sides by 8: $x = 48$

2.4 Solving Multi-Step Equations

In this section, we'll learn how to solve equations with more than one operation. We'll use combining like terms and distribution before solving. First, **simplify both sides** of the equation using distribution and combining like terms. Then, **isolate the variable** step by step using inverse operations.

Solving multi-step equations builds confidence and logical thinking. You break down a big problem into smaller pieces—just like solving a mystery one clue at a time. People use multi-step equations in budgeting, engineering, and even cooking recipes when they need to scale quantities.

Example: Multi-Step Problem Solving

Let's solve the equation $2(x + 3) = 14$ step by step:

Step 1: Distribute

$$2(x + 3) = 14 \quad \rightarrow \quad 2x + 6 = 14$$

Step 2: Subtract 6 from both sides

$$2x + 6 - 6 = 14 - 6 \quad \rightarrow \quad 2x = 8$$

Step 3: Divide both sides by 2

$$\frac{2x}{2} = \frac{8}{2} \quad \rightarrow \quad x = 4$$

Don't rush—write each step clearly on its own line. Make sure each side of the equation is simplified before you solve. Always check your solution by plugging it back into the original equation.

Short Question (English)

Solve: $2x + 5 = 13$

Short Question (Korean)

Solve: $2x + 5 = 13$

Short Question (Answer)

Solve: $2x + 5 = 13$

Answer: $2x = 8 \rightarrow x = 4$

Long Question (English)

Solve: $3(x - 2) = 15$

Long Question (Korean)

Solve: $3(x - 2) = 15$

Long Question (Answer)

Solve: $3(x - 2) = 15$

Answer: $x - 2 = 5 \rightarrow x = 7$

Challenge Question (English)

Solve: $4x + 7 = 3x + 12$

Challenge Question (Korean)

Solve: $4x + 7 = 3x + 12$

Challenge Question (Answer)

Solve: $4x + 7 = 3x + 12$

Answer:

Subtract $3x$: $x + 7 = 12$

Subtract 7: $x = 5$

2.5 Inequalities

In this section, we'll learn how to solve basic inequalities and how to graph your answers on a number line. Inequalities use signs like:

- $>$ greater than
- $<$ less than
- $\geq$ greater than or equal to
- $\leq$ less than or equal to

Solve inequalities just like equations, **but remember**: If you multiply or divide both sides by a **negative number**, you must **flip the inequality sign**.

Inequalities are used in many jobs—like making decisions in business, writing computer programs, or planning safe speeds in transportation.

Example: Solving Inequalities

Let's look at some examples:

Basic Inequality:

$$x - 3 > 2 \quad \text{Add 3 to both sides} \quad x > 5$$

Inequality with Negative Coefficient:

$$-2x < 6 \quad \text{Divide both sides by -2, flip the sign} \quad x > -3$$

Suppose a sign says, "You must be **under 18** to enter." That's an inequality! It means your age must be **less than 18**, or $x < 18$. Inequalities describe rules, limits, and ranges all around us.

Always treat inequalities like equations—unless you multiply or divide by a negative! Graph solutions using number lines. Use:

- **Open circles** for $<$ or $>$
- **Closed circles** for $\leq$ or $\geq$

Check your solution by testing a number that fits the inequality.

Short Question (English)

Solve: $x + 4 < 10$

Short Question (Korean)

Solve: $x + 4 < 10$

Short Question (Answer)

Solve: $x + 4 < 10$

Answer: $x < 6$

Long Question (English)

Solve and graph: $-3x \geq 12$

Long Question (Korean)

Solve and graph: $-3x \geq 12$

Long Question (Answer)

Solve and graph: $-3x \geq 12$

Answer:

Divide by -3 and flip the sign: $x \leq -4$

Graph: closed circle at -4 , shade to the left.

Challenge Question (English)

A movie ticket costs \$12. Write and solve an inequality for the number of tickets you can buy with \$80.

Challenge Question (Korean)

A movie ticket costs \$12. Write and solve an inequality for the number of tickets you can buy with \$80.

Challenge Question (Answer)

A movie ticket costs \$12. Write and solve an inequality for the number of tickets you can buy with \$80.

Answer:

$$12t \leq 80$$

$t \leq 6.\overline{6} \rightarrow$ You can buy at most **6 tickets**.

2.6 Summary and Tips

Interesting Facts

- The word **"algebra"** comes from the Arabic term **"al-jabr,"** which means "reunion of broken parts." It appeared in a famous math book written over 1,200 years ago by a scholar named Al-Khwarizmi. Many English math words like "algorithm" also come from his name!
- Even NASA engineers and video game developers simplify expressions when building models and writing code. It's a skill that goes far beyond school!
- Solving equations is a lot like solving a riddle or puzzle. You're looking for the one number that makes the equation true.
- People use multi-step equations in budgeting, engineering, and even cooking recipes when they need to scale quantities.
- Inequalities are used in many jobs—like making decisions in business, writing computer programs, or planning safe speeds in transportation.

Tips for Students

Variables and Algebraic Expressions

- Think of a variable as a mystery box. What number could fit inside?
- Try different values to see how expressions change.
- When writing expressions, use colors, arrows, or circles to match terms and operations.

Simplifying Expressions

- Circle or highlight like terms before you combine them.
- Be careful with minus signs—they belong to the number or variable that follows!
- Go one step at a time to avoid careless mistakes.

Solving One-Step Equations

- Ask: "What's happening to the variable?" Then do the opposite.
- Keep both sides of the equation equal—whatever you do to one side, do to the other.
- Always plug your answer back in to make sure it works!

Solving Multi-Step Equations

- Don't rush—write each step clearly on its own line.
- Make sure each side of the equation is simplified before you solve.
- Always check your solution by plugging it back into the original equation.

Inequalities

- Always treat inequalities like equations—unless you multiply or divide by a negative!
- Graph solutions using number lines. Use:
 - **Open circles** for $<$ or $>$
 - **Closed circles** for $\leq$ or $\geq$
- Check your solution by testing a number that fits the inequality.

Chapter 3

Ratios, Rates, and Percents

This is an example sentence with an index entry. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum.

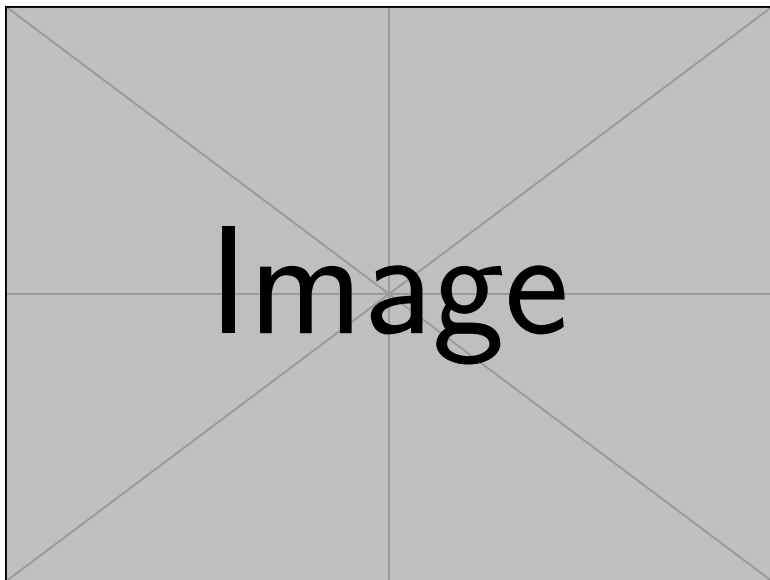


Figure 3.1: A scenic travel destination (example image).

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3.1 Ratios

Welcome to Unit 3! In this section, we'll learn how to compare amounts using ratios. A **ratio** is a comparison between two numbers or amounts. It tells us how much of one thing there is compared to another. Ratios help us describe relationships and patterns, and they appear in recipes, sports, and even classrooms.

Example: If there are 2 apples and 3 oranges, the ratio of apples to oranges is **2 to 3**.

Ways to Write Ratios

There are several ways to write ratios:

- With a colon: **2:3**
- As a fraction: **$\frac{2}{3}$**
- In words: **2 to 3**

All three ways mean the same thing.

Part-to-Part and Part-to-Whole

- **Part-to-Part:** Compares two individual parts (e.g., apples to oranges $\rightarrow$ 2:3).
- **Part-to-Whole:** Compares one part to the total (e.g., apples to total fruits $\rightarrow$ 2:5).

Example: If a class has 12 boys and 8 girls:

- The ratio of boys to girls is **12:8** $\rightarrow$ simplified to **3:2**.
- The ratio of girls to total students (20) is **8:20** $\rightarrow$ simplified to **2:5**.

Ratios are used to mix paint colors, create scale models, adjust ingredients in recipes, and describe odds in games!

Short Question (English)

There are 6 red marbles and 4 blue marbles in a bag. What is the ratio of red marbles to blue marbles?

Short Question (Korean)

There are 6 red marbles and 4 blue marbles in a bag. What is the ratio of red marbles to blue marbles?

Short Question (Answer)

There are 6 red marbles and 4 blue marbles in a bag. What is the ratio of red marbles to blue marbles?

Answer: **6:4**, which can be simplified to **3:2**.

Long Question (English)

A class has 12 boys and 8 girls.

- a) Write the ratio of boys to girls in simplest form.
- b) Write the ratio of girls to total students in simplest form.

Long Question (Korean)

A class has 12 boys and 8 girls.

- a) Write the ratio of boys to girls in simplest form.
- b) Write the ratio of girls to total students in simplest form.

Long Question (Answer)

A class has 12 boys and 8 girls.

- a) Write the ratio of boys to girls in simplest form.
- b) Write the ratio of girls to total students in simplest form.

Answer:

- a) Boys to girls = **12:8**, simplified to **3:2**.
- b) Girls to total = $8 : 20$, simplified to **2:5**.

Challenge Question (English)

A recipe for bibimbap calls for 2 cups of rice and 3 cups of vegetables. If you want to make enough for twice as many people, keeping the same ratio, how many cups of rice and vegetables do you need?

Challenge Question (Korean)

A recipe for bibimbap calls for 2 cups of rice and 3 cups of vegetables. If you want to make enough for twice as many people, keeping the same ratio, how many cups of rice and vegetables do you need?

Challenge Question (Answer)

A recipe for bibimbap calls for 2 cups of rice and 3 cups of vegetables. If you want to make enough for twice as many people, keeping the same ratio, how many cups of rice and vegetables do you need?

Answer:

Rice: $2 \times 2 = 4$ cups

Vegetables: $3 \times 2 = 6$ cups

3.2 Rates

A **rate** is a special type of ratio that compares two different kinds of units. For example, it can compare miles to hours, dollars to items, or words to minutes.

Example: If you drive 100 kilometers in 2 hours, the rate is:

- 100 km per 2 hours $\rightarrow$ **50 km/h** (kilometers per hour)

Common Rates

- **Speed:** miles per hour (mph), kilometers per hour (km/h)
- **Prices:** dollars per item (\$/item)
- **Typing:** words per minute (wpm)

Everyday Examples

- A printer prints 60 pages in 3 minutes $\rightarrow$ Rate: **20 pages per minute**
- A worker earns \$200 for 8 hours $\rightarrow$ Rate: **\$25/hour**

Rates are everywhere—in recipes, gas mileage (MPG), data usage on your phone, and how fast athletes run.

Short Question (English)

A car travels 180 kilometers in 3 hours. What is its speed in km/h?

Short Question (Korean)

A car travels 180 kilometers in 3 hours. What is its speed in km/h?

Short Question (Answer)

A car travels 180 kilometers in 3 hours. What is its speed in km/h?

Answer: $180 \div 3 = 60$ km/h

Long Question (English)

A market sells 5 kg of rice for \$20. What is the rate in dollars per kilogram?

Long Question (Korean)

A market sells 5 kg of rice for \$20. What is the rate in dollars per kilogram?

Long Question (Answer)

A market sells 5 kg of rice for \$20. What is the rate in dollars per kilogram?

Answer: $20 \div 5 = 4 \rightarrow$ **\$4 per kg**

Challenge Question (English)

A Korean drama episode is 75 minutes long, and you watch 3 episodes in a day. How many minutes per day do you watch? Express as a rate in hours per day.

Challenge Question (Korean)

A Korean drama episode is 75 minutes long, and you watch 3 episodes in a day. How many minutes per day do you watch? Express as a rate in hours per day.

Challenge Question (Answer)

A Korean drama episode is 75 minutes long, and you watch 3 episodes in a day. How many minutes per day do you watch? Express as a rate in hours per day.

Answer:

Total minutes: $75 \times 3 = 225$ minutes

Convert to hours: $225 \div 60 = 3.75$ hours/day

3.3 Unit Rate

A **unit rate** is a rate where the second quantity is exactly **1**. It tells you how much of something there is **per one unit** of something else.

****Example:**** If 3 notebooks cost \$6, the unit rate is:

- $\$6 \div 3 = \text{\$2 per notebook}$

How to Find a Unit Rate

1. Divide the first number by the second.
2. Make sure the second number becomes 1.

****Examples:****

- 120 miles in 4 hours $\rightarrow 120 \div 4 = 30$ miles per hour
- 8 apples for \$4 $\rightarrow \$4 \div 8 = \0.50 per apple

Real-Life Uses

- Grocery stores often list unit prices on shelf tags to help shoppers compare.
- Runners track their pace (minutes per mile) to measure speed.
- Restaurants calculate price per serving to set menu prices.

Unit rates are super helpful when budgeting, comparing phone plans, or deciding which product gives you the best deal!

Short Question (English)

If 4 notebooks cost \$10, what is the cost per notebook?

Short Question (Korean)

If 4 notebooks cost \$10, what is the cost per notebook?

Short Question (Answer)

If 4 notebooks cost \$10, what is the cost per notebook?

Answer: $10 \div 4 = 2.50 \rightarrow$ **\$2.50 per notebook**

Long Question (English)

A package of 12 bottles of water costs \$6. Find the unit price per bottle.

Long Question (Korean)

A package of 12 bottles of water costs \$6. Find the unit price per bottle.

Long Question (Answer)

A package of 12 bottles of water costs \$6. Find the unit price per bottle.

Answer: $6 \div 12 = 0.50 \rightarrow \0.50 per bottle

Challenge Question (English)

At a Korean grocery store, a 2-pound bag of tangerines costs \$5.50, and a 3-pound bag costs \$7.20. Which is the better deal per pound?

Challenge Question (Korean)

At a Korean grocery store, a 2-pound bag of tangerines costs \$5.50, and a 3-pound bag costs \$7.20. Which is the better deal per pound?

Challenge Question (Answer)

At a Korean grocery store, a 2-pound bag of tangerines costs \$5.50, and a 3-pound bag costs \$7.20. Which is the better deal per pound?

Answer:

2-pound bag: $5.50 \div 2 = 2.75$ \$/lb

3-pound bag: $7.20 \div 3 = 2.40$ \$/lb

Better deal: 3-pound bag (\$2.40/lb)

3.4 Proportions

A **proportion** is a statement that shows two ratios or rates are **equal**. It helps us solve problems where quantities grow or shrink at the same rate.

****Example:**** If $2/3 = 4/6$, this is a proportion because both sides are equal.

How to Solve Proportions

Use **cross multiplication**:

- $2/3 = 4/6$
- $2 \cot 6 = 3 \cot 4$
- $12 = 12$. This means it is a true proportion.

You can also use proportions to find unknown numbers:

****Example:****

$$\frac{x}{5} = \frac{6}{10} \Rightarrow 10x = 30 \Rightarrow x = 3$$

Everyday Examples

- A map says 1 inch = 50 miles $\rightarrow$ This ratio helps calculate real distances.
- A recipe that serves 2 people can be doubled to serve 4 using proportions.

Proportions are used in architecture, photography, and fashion design to maintain balance and scale!

Short Question (English)

Is $2/5 = 8/20$ a true proportion?

Short Question (Korean)

Is $2/5 = 8/20$ a true proportion?

Short Question (Answer)

Is $2/5 = 8/20$ a true proportion?

Answer: Yes. Both simplify to $2/5$.

Long Question (English)

If 3 cups of flour are needed to make 12 cookies, how many cups of flour are needed to make 20 cookies, assuming the recipe stays in proportion?

Long Question (Korean)

If 3 cups of flour are needed to make 12 cookies, how many cups of flour are needed to make 20 cookies, assuming the recipe stays in proportion?

Long Question (Answer)

If 3 cups of flour are needed to make 12 cookies, how many cups of flour are needed to make 20 cookies, assuming the recipe stays in proportion?

Answer:

Set up: $3/12 = x/20$

Cross-multiply: $12x = 60 \rightarrow x = 5$ cups

Challenge Question (English)

A map uses a scale of $1 \text{ cm} = 5 \text{ km}$. If two cities are 8.4 cm apart on the map, how far apart are they in real life?

Challenge Question (Korean)

A map uses a scale of $1 \text{ cm} = 5 \text{ km}$. If two cities are 8.4 cm apart on the map, how far apart are they in real life?

Challenge Question (Answer)

A map uses a scale of $1 \text{ cm} = 5 \text{ km}$. If two cities are 8.4 cm apart on the map, how far apart are they in real life?

Answer: $8.4 \times 5 = 42 \text{ km}$

3.5 Percent

A **percent** is a number that means "per 100." It shows a part out of a whole divided into 100 equal parts.

****Examples:****

- 50% means 50 out of 100 $\rightarrow$ or $\frac{1}{2}$
- 25% means 25 out of 100 $\rightarrow$ or $\frac{1}{4}$
- 100% means all of something

How to Convert

- **Fraction to Percent:** Multiply by 100 $\frac{1}{4} = 0.25 \Rightarrow 25\%$
- **Decimal to Percent:** Move the decimal point two places to the right $0.6 \Rightarrow 60\%$
- **Percent to Decimal:** Move the decimal two places left $45\% = 0.45$

Real-Life Examples

- **Shopping:** 30% off a jacket means you save 30 out of every 100 dollars.
- **Grades:** A score of 85% means you got 85 out of 100 points.
- **Interest Rates:** Banks use percents to calculate how much money you earn or owe.

The symbol % was first used hundreds of years ago in Europe. Percentages are now used globally in business, science, and even sports statistics!

Short Question (English)

Write 0.45 as a percent.

Short Question (Korean)

Write 0.45 as a percent.

Short Question (Answer)

Write 0.45 as a percent.

Answer: Move decimal two places right: $0.45 = 45\%$

Long Question (English)

A store is having a 25% off sale. If a jacket costs \$60 before the sale, how much will it cost after the discount?

Long Question (Korean)

A store is having a 25% off sale. If a jacket costs \$60 before the sale, how much will it cost after the discount?

Long Question (Answer)

A store is having a 25% off sale. If a jacket costs \$60 before the sale, how much will it cost after the discount?

Answer:

$$25\% \text{ of } \$60 = 0.25 \times 60 = 15$$

$$\text{Discounted price} = 60 - 15 = 45$$

Answer: \$45

Challenge Question (English)

In a class of 40 students, 70% passed an exam. How many students passed, and how many failed?

Challenge Question (Korean)

In a class of 40 students, 70% passed an exam. How many students passed, and how many failed?

Challenge Question (Answer)

In a class of 40 students, 70% passed an exam. How many students passed, and how many failed?

Answer:

Passed: $0.70 \times 40 = 28$ students

Failed: $40 - 28 = 12$ students

3.6 Summary and Tips

Interesting Facts

- Ratios are used to mix paint colors, create scale models, adjust ingredients in recipes, and describe odds in games!
- Rates are everywhere—in recipes, gas mileage (MPG), data usage on your phone, and how fast athletes run.
- Unit rates are super helpful when budgeting, comparing phone plans, or deciding which product gives you the best deal!
- Proportions are used in architecture, photography, and fashion design to maintain balance and scale!
- The symbol % was first used hundreds of years ago in Europe. Percentages are now used globally in business, science, and even sports statistics!

Tips for Students

Ratios

- Keep the order of comparison the same—read carefully.
- Simplify ratios when possible, just like you simplify fractions.
- Think visually—use pictures or charts if it helps you understand.

Rates

- Always include the units (like miles, hours, or dollars).
- To find a rate, divide the first number by the second.
- Use rates to compare which deal is better when shopping.

Unit Rate

- Make sure the second number becomes 1 (that's what "unit" means!).
- Use real-world examples when practicing—like prices, distance, or time.
- Always include the unit in your answer (like "per hour" or "per item").

Proportions

- Always simplify ratios before comparing.
- Use cross multiplication to solve quickly.
- Write word problems as proportions to organize your thinking.

Percent

- Think of % as "out of 100."
- Use visual aids like 100 grids or pie charts to understand percent.
- Practice converting between fractions, decimals, and percents until it feels natural.

- The word “digit ‘digit2’ is the one!” come ...
- The “word” “digit” come ...

“To be, or not to be, that is the question: Whether’s tis nobler in the mind to suffer. The slings and arrows of outrageous fortune...” (Kim 2003).

“This is your quoted text.” (Kim 2003, p. 359).

Tip for Students

Example sentence is here. (Kim 2003, p. 359).

“Example sentence is here.” The author said this (*Help your kids with math* 2010, p. 200).

Appendix

Important Formulas and Definitions

- **Place Value:** Value of a digit = digit $\times$ place value.
- **Fraction to Decimal:** $\frac{a}{b} = a \div b$.
- **Prime Number:** A number greater than 1 with exactly two factors: 1 and itself.
- **Composite Number:** A number with more than two factors.
- **GCF:** Product of lowest powers of common primes.
- **LCM:** Product of highest powers of all primes present.

Further Reading and Resources

- "Elementary Mathematics" by Liping Ma
- "How to Be a Math Genius" by Mike Goldsmith
- Khan Academy: <https://www.khanacademy.org/>
- Purplemath: <https://www.purplemath.com/>

Summary of Key Concepts

Topic	Key Points
Place Value	Each digit's value depends on its position. Example: $4275 = 4000 + 200 + 70 + 5$.
Decimals and Fractions	Decimals and fractions both represent parts of a whole. Example: $0.75 = \frac{3}{4}$.
Comparing and Ordering Numbers	Use $>$, $<$, $=$ to compare. Convert to decimals for easy comparison. Example: $\frac{2}{5} < \frac{3}{7}$.
Factors and Multiples	Factors divide a number evenly; multiples are products of a number and an integer. Example: Factors of 12 are 1, 2, 3, 4, 6, 12.
Prime and Composite Numbers	Primes have exactly two factors (1 and itself); composites have more. Example: 2, 3, 5 are prime; 4, 6, 8 are composite.
GCF and LCM	GCF: largest common factor; LCM: smallest common multiple. Use prime factorization. Example: GCF of 12 and 18 is 6; LCM is 36.

Table 3.1: **Summary of Key Concepts**

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- [2] *Help your kids with math: a unique step-by-step visual guide*. First American edition. New York, New York: DK Publishing, 2010. ISBN: 9780756649791.